Classifying Heart Disease from BRFSS Data

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***Abstract*—We compare least squares, ridge, and logistic regression for classifying a reported history of myocardial infarction or coronary heart disease from the 2015 BRFSS survey. The training data has 328,135 respondents, of whom 8.83% belong to the positive class. Before fitting each model, we handle coded and missing answers and then evaluate each model using stratified five-fold cross-validation to preserve original class proportions within each fold. Throughout our model evaluations, we use decision thresholds selected on the inner training splits. Our selected unregularized logistic regression model achieves a mean validation F_1 of 0.4275 (standard deviation 0.0031) and a test (AIcrowd) F_1 score of 0.439. The biggest performance gains came from correctly interpreting survey answers and choosing the best decision thresholds. On the other hand, regularization and class rebalancing provided no clear F1 improvement.**

***Index Terms*—Heart disease, BRFSS, Logistic regression, Cross-validation, Class imbalance**

I. Introduction

In 2015, the Behavioral Risk Factor Surveillance System (BRFSS) collected responses about demographics, lifestyle, and health. In our study, we investigate whether these survey responses can help identify people who report a history of myocardial infarction or coronary heart disease. To do this, we compare different regression-based classification methods, examining how preprocessing and model evaluation choices affect their performance.

Initial data exploration revealed that accuracy alone is a poor guide to performance because of class imbalance. Predicting every respondent as negative would achieve 91.17% accuracy without detecting a single positive case. We therefore focus on $F_1 = \frac{2PR}{P+R}$, where P denotes precision and R denotes recall.

Some features also contain missing answers and numeric codes that require different treatment. For example, "none" represents a valid zero, while "don't know" is treated as missing information. Treating both as ordinary numbers would be a mistake, changing the meaning of the responses.

II. Methods

The survey is not a clean numeric table, and the official codebook [1] is essential to read it. Our pipeline (in `preprocessing.py`) applies five steps.

Excluding columns (103 of 321). We drop 31 administrative variables (identifiers, dates, interview mode, survey weights), 2 date columns, and the state code. We also drop **one leakage column**: `HAREHAB1` (cardiac rehabilitation) is only asked to people who reported a heart attack, which is part of the definition of the target. Finally we drop 68 **redundant** columns that the codebook defines as exact functions of other columns.

Recoding special codes. In BRFSS, "don't know" and "refused" are stored as 7, 9, 77, 99, 777 or 999, and "none"/"never" as 88, 555 or 888. We map the first group to missing and the second to 0, with the rule chosen by the range of each column and a list of exceptions checked one by one in the codebook (77 is a valid number of drinks only up to 76; 88 is a valid age at diabetes diagnosis).

Handling missing values. Every respondent has at least 25% missing answers (median 43%), so rows cannot be dropped; we work column-wise. Most missingness is not random: it comes from **skip patterns**, questions asked only to a subgroup (blood-pressure medication only to hypertensive people). We keep the columns with at most a fraction τ of missing values, impute the median, and add one binary indicator per distinct missingness pattern. We selected τ over the grid $0.1, 0.3, 0.5, 0.8, 0.9$ by cross-validation (F1 from 0.4133 to 0.4258, AUC from 0.8517 to 0.8589, both increasing); $\tau = 0.9$ was best but within 0.6 SE of $\tau = 0.8$ on F1, so by the rule below we kept the smaller, simpler model.

Encoding and scaling. For 23 variables the numeric code carries no order (marital status) or an order that does not match the risk (`_SMOKER3`: former smokers are at higher risk than current ones), so we one-hot encode them, using $k-1$ dummies to avoid collinearity with the bias. All features are then standardised with the training statistics.

With $\tau = 0.8$ this turns 106 survey questions into **277 features** (83 numeric, 95 dummies, 98 missingness indicators and the bias).

The decision threshold is itself a hyperparameter, to select it we used **stratified 5-fold cross-validation** (seed 42). In each fold we pick the F1-maximising threshold on an inner fifth of the training part, then refit on the whole training part and evaluate on the held-out fold, so the evaluation data never influence any choice.

To compare two configurations we use their **paired** per-fold differences, which cancel the variability due to some folds being easier than others. The five folds are not independent (their training sets overlap by 75%), so we correct the standard error [2]:

$$\widehat{\mathrm{SE}} = \sqrt{\left(\frac{1}{k} + \frac{n_{\text{test}}}{n_{\text{train}}}\right) s_d^2}, \quad \frac{n_{\text{test}}}{n_{\text{train}}} = \frac{1}{k-1}. \tag{1}$$

We also report AUC, the probability that a random patient is ranked above a random non-patient. AUC does not depend on the threshold and is much less noisy than F1, which makes it the better detector of small real improvements. **All decision rules were fixed before looking at the numbers:** a modification is adopted only if it improves F1 by at least 2 SE **and** 0.002, without lowering AUC by more than 2 SE.

All six required methods share the same folds and pipeline. Iterative methods start from $w = 0$ and use the step size $\gamma = 1/L$ given by the Lipschitz constant of the gradient, $L = \lambda_{\max}(X^\top X/n)/4$ for the logistic loss, which guarantees that the loss decreases at every step. We use 1000 iterations: doubling them changes the validation AUC by only $3 \cdot 10^{-4}$.

III. Results

TABLE I
Cross-validated performance.

Model	F1	AUC
Always positive	0.162	0.500
Logistic (age, sex only)	0.294	0.753
Least squares (GD/SGD converge to it)	0.4140	0.8518
Ridge, $\lambda = 0.1$	0.4142	0.8529
Logistic	**0.4275**	**0.8600**
Regularized logistic, $\lambda = 10^{-4}$	0.4271	0.8600

From the results we see that logistic loss beats squared loss and regularization is unnecessary. Logistic regression outperforms least squares by $+0.0135$ F_1 and $+0.0082$ AUC, see Table I. First, disease risk in our data grows linearly on the logit scale (especially with age), which matches the logistic assumption. Second, squared loss penalizes "too confident" correct predictions, causing the 91% negative class to dominate the fit, whereas logistic loss focuses on the decision boundary. By contrast, L_2 regularization brings no real benefit: ridge improves least-squares AUC by just $+0.0011$, and regularized logistic regression matches the unregularized model. With $262,508$ training rows for only 277 parameters, overfitting is already negligible.

Giving positive samples higher weight ($\alpha \in \{2, 5, 10.3\}$) or undersampling negatives ($r = 1/\alpha$) never improved F_1. In logistic regression, weighting positives by α simply shifts the log-odds by $\log \alpha$. This leaves the patient ranking (AUC) unchanged, and our tuned threshold automatically shifts to cancel out the weight (from $t = 0.196$ up to $0.333, 0.554$, and 0.734, matching the theoretical values $0.328, 0.549$, and 0.715). Undersampling shifts the threshold the same way, but throwing away data increases variance and hurts AUC.

Training on growing subsets of data (from 5% to 100%) shrinks the train–validation AUC gap to zero as both curves plateau at ~ 0.861. Since the linear model has hit its ceiling, we tested non-linear features. Quintile-binning continuous variables changed nothing ($\Delta\text{AUC} = +0.0001$). Interacting all features with age (552 features) improved the ranking ($+0.0014$ AUC, 4.9 SE) because risk factors vary with age, but the F_1 gain ($+0.0010$, 0.6 SE) was too small to justify doubling the model size. Sex interactions added nothing. This means that our model is limited by bias, not variance.

TABLE II
Ablation: one component removed at a time from the final model. All differences except the last F1 are significant at more than 2 SE.

Component removed	ΔF1	ΔAUC
Tuned threshold (0.5 instead)	-0.1939	0
Recoding of special codes	-0.0173	-0.0111
Logistic loss (least squares)	-0.0135	-0.0082
One-hot encoding	-0.0025	-0.0013
$\tau = 0.8$ ($\tau = 0.3$ instead)	-0.0020	-0.0007
Missingness indicators	-0.0000	-0.0005

Removing one step at a time (Table II) shows that tuning the threshold is critical: using $t = 0.5$ maximizes accuracy (0.915) but ruins F_1 (0.2336). Our tuned threshold ($t = 0.196$) matches the theoretical optimum $F_1^*/2$ [3], effectively treating a missed patient as four times worse than a false alarm. Second is recoding special survey codes (-0.0173 F_1), which matters more than the choice of ML model itself.

IV. Conclusion

In conclusion, we retain unregularized logistic regression with a submission threshold of $t = 0.20$, achieving a test (AIcrowd) F_1 score of 0.439. The largest changes in performance came from threshold selection and the transformation of survey codes into usable numbers. Having the threshold at 0.5 reduces mean validation F1 from 0.4275 to 0.2336, while omitting the recoding step reduces it to 0.4102. We saw that regularization, class weighting, and undersampling provide no clear F1 gains. Finally, interactions with age improve AUC, but their F1 improvement does not meet our selection criterion. We therefore keep the simpler model.

Reproducibility. `python run.py` with the dataset in `dataset/` rebuilds the submission (seed 42, no randomness other than the fold split). The notebook `project1_notebook.ipynb` contains every experiment reported here; all of them import the same `preprocessing.py` and `validation.py`.

References

[1] Centers for Disease Control and Prevention, "BRFSS 2015 Codebook Report," technical report, 2016.

[2] C. Nadeau and Y. Bengio, "Inference for the Generalization Error," *Machine Learning*, vol. 52, no. 3, 2003.

[3] Z. C. Lipton, C. Elkan, and B. Narayanaswamy, "Optimal Thresholding of Classifiers to Maximize F1 Measure," 2014.